

MAT 528 HW06

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Due 10/09

Problem 1 (Munkres 26.3). Show that a finite union of compact subspaces of X is compact.

Solution. Let $C_1, \dots, C_n$ be compact subspaces of X . If $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ is an open cover of $C_1 \cup \dots \cup C_n$, then it is an open cover of each C_k for $k = 1, \dots, n$. Since C_k is compact, it has a finite subcover $\mathcal{U}_k \subseteq \mathcal{U}$, so $C_1 \cup \dots \cup C_n$ has the finite open cover $\mathcal{U}_1 \cup \dots \cup \mathcal{U}_n \subseteq \mathcal{U}$. Thus, given any open cover of $C_1 \cup \dots \cup C_n$ we can extract a finite subcover, which is what it means for $C_1 \cup \dots \cup C_n$ to be compact. $\square$

Problem 2 (Munkres 26.4). Show that every compact subspace of a metric space is bounded in that metric space. Find a metric space in which not every closed bounded space is compact.

Solution. Suppose C is a compact subspace of metric space X with metric d . Then, fixing $\varepsilon > 0$, C has an open cover $\{B_\varepsilon(x)\}_{x \in C}$ consisting of balls centered at each $x \in C$, and this open cover has a finite subcover $\mathcal{U} = \{B_\varepsilon(x_j) : x_1, \dots, x_n \in C\}$ of open balls corresponding to some finitely many points $x_1, \dots, x_n \in C$.

If $x_0 \in X$ is an arbitrary fixed point, then for any $y \in X$, $B_{d(x_0, y) + \varepsilon}(x_0)$ contains $B_\varepsilon(y)$ because if $y' \in B_\varepsilon(y)$ then

$$d(x_0, y') \leq d(x_0, y) + d(y, y') < \varepsilon.$$

This means that for $k = 1, \dots, n$, $B_\varepsilon(x_k) \in \mathcal{U}$ is contained in $B_{d(x_0, x_k) + \varepsilon}(x_0)$, so if $M = \max\{d(x_0, x_k) : k = 1, \dots, n\}$ then

$$C \subseteq \bigcup_{k=1}^n B_\varepsilon(x_k) = B_{M+\varepsilon}(x_0),$$

so C is bounded because it is contained in an open ball. (more specifically, C is bounded because $d(x, y) < 2(M + \varepsilon)$ for any $x, y \in B_{M+\varepsilon}(x_0)$, hence the same inequality holds for any $x, y \in C$, so C has diameter at most $2(M + \varepsilon)$)

On the other hand, give $\mathbb{R}$ the metric

$$\begin{aligned} d(x, y) &= 1 \text{ if } x \neq y \\ d(x, y) &= 0 \text{ if } x = y. \end{aligned}$$

Then $\mathbb{R}$ is closed (by definition) and bounded (because $d(x, y) \leq 1$ for all $x, y \in \mathbb{R}$) but $\mathbb{R}$ is not compact because, by example 1 of section 20 in Munkres, d induces the discrete topology, so the collection of singleton sets $\{\{x\}\}_{x \in \mathbb{R}}$ is an open cover, which cannot have a finite subcover since the union of any finite subcollection of these singleton sets is a finite set, and $\mathbb{R}$ is not finite. $\square$

Problem 3 (Munkres 26.7). Show that if Y is compact, then the projection $\pi_1 : X \times Y \rightarrow X$ is a closed map.

Solution. Let $F \subseteq X \times Y$ be a closed set. We show that the complement of $\pi_1(F)$ is open. Let $x_0 \in X \setminus \pi_1(F)$; to show that $X \setminus \pi_1(F)$ we need to show that there exists an open set W such that $x_0 \in W \subseteq X \setminus \pi_1(F)$. The set $(X \times Y) \setminus F$ is an open set containing $\{x_0\} \times Y$, so by the tube lemma $(X \times Y) \setminus F$ contains a tube $W \times Y$ where $W \subseteq X$ is an open set containing x_0 . In particular, no point of $W \times Y$ is in F , so $\pi_1(W) \subseteq X \setminus \pi_1(F)$ is an open set in X (because π_1 is an open map, as shown on a previous homework) that contains x_0 ; this proves that $X \setminus \pi_1(F)$ is open. $\square$

Problem 4 (Munkres 26.8). Let Y be a compact Hausdorff space and let $f: X \rightarrow Y$. Then f is continuous if and only if the **graph** G_f of f ,

$$G_f := \{(x, f(x)) : x \in X\},$$

is closed in $X \times Y$.

Solution. Suppose f is continuous and let (x_0, y_0) be a point in $(X \times Y) \setminus G_f$. Then $f(x_0) \neq y_0$, so there exist disjoint open sets $U, V \subseteq Y$ such that $f(x_0) \in U$ and $y_0 \in V$ because Y is Hausdorff. Then, the open set $f^{-1}(U) \times V$ is disjoint from G_f because V does not contain any points in $f(f^{-1}(U)) = U$, which is enough to prove that the complement of G_f is open in $X \times Y$, hence G_f is closed.

Conversely suppose that G_f is closed in $X \times Y$. Let V be an open set in Y and let $F = (X \times V) \cap G_f$, which is closed. Then F is the set

$$\{(x, y) \in G_f : y \in V\} = \{(x, f(x)) \in X \times Y : x \in X \text{ such that } f(x) \in V\},$$

so $\pi_1(F) = f^{-1}(V)$ because it is the set of all $x \in X$ for which $f(x) \in V$; since Y is compact, the preceding problem implies that π_1 is a closed map, so $\pi_1(F) = f^{-1}(V)$ is closed for every closed set F , implying f continuous. $\square$

Problem 5 (Munkres 27.2ab). Let X be a metric space with metric d and let $A \subseteq X$ be nonempty.

- (a) Show that $d(x, A) = 0$ if and only if $x \in \overline{A}$.
- (b) Show that if A is compact, $d(x, A) = d(x, a)$ for some $a \in A$.

Solution. (a) If $x \in \overline{A}$ then for each $\varepsilon > 0$ the ball $B_\varepsilon(x)$ contains some point $y \in A$, so $d(x, A) \leq d(x, y) < \varepsilon$; this implies $d(x, A) = 0$.

Conversely, if $x \notin \overline{A}$ then there exists $\varepsilon > 0$ such that $B_\varepsilon(x)$ does not intersect A , meaning for any $y \in A$, $d(x, y) \geq \varepsilon$; taking the infimum over all $y \in A$ gives $d(x, A) \geq \varepsilon > 0$.

- (b) The function $d(x, -): X \rightarrow \mathbb{R}$ is continuous at each $y \in X$ because for each $\varepsilon > 0$,

$$d(y, y') < \varepsilon \implies |d(x, y) - d(x, y')| \leq d(y, y') < \varepsilon$$

where $|d(x, y) - d(x, y')| \leq d(y, y')$ because $d(x, y) \leq d(x, y') + d(y, y')$ and $d(x, y') \leq d(x, y) + d(y', y)$. Therefore by Munkres's Theorem 27.4, since A is compact, $d(x, -)$ there exists a point $a \in A$ for which

$$d(x, a) = \inf_{y \in A} d(x, y) = d(x, A). \quad \square$$

Problem 6 (Munkres 27.6). Let A_0 be the closed interval $[0, 1]$ in $\mathbb{R}$. Let A_1 be the set obtained from A_0 by deleting its “middle third” $(\frac{1}{3}, \frac{2}{3})$. Let A_2 be the set obtained from A_1 by deleting its “middle thirds” $(\frac{1}{9}, \frac{2}{9})$ and $(\frac{7}{9}, \frac{8}{9})$. In general, define A_n by the equation

$$A_n = A_{n-1} \setminus \bigcup_{k=0}^{\infty} \left(\frac{1+3k}{3^n}, \frac{2+3k}{3^n} \right).$$

The intersection $C = \bigcap_{n=1}^{\infty} A_n$ is called the **Cantor set**; it is a subspace of $[0, 1]$.

- (a) Show that C is totally disconnected.
- (b) Show that C is compact.
- (c) Show that each set A_n is a union of finitely many disjoint closed intervals of length 3^{-n} , and show that the endpoints of these intervals lie in C .

(d) Show that C has no isolated points.

(e) Conclude that C is uncountable.

Solution. (a) Suppose $A \subseteq C$ contains two distinct points x, y . Then, pick n so that $3^{-n} < |x - y|$, so that by (c), x and y must lie in distinct intervals of A_n . If I is the interval containing x then I and $A_n \setminus I$ are both closed in A_n because they are unions of finitely many closed intervals, nonempty, and disjoint, so $I \cap A$ and $A \cap (A_n \setminus I) = A \setminus I$ are closed in I , nonempty (since I contains x but not y so $A \setminus I$ contains y), and disjoint, which means we have found a separation of A . Thus C is totally disconnected because we have shown that any set $A \subseteq C$ with more than two elements is disconnected.

(b) Each A_n is closed because it is the intersection of closed sets

$$A_n = A_{n-1} \setminus \bigcup_{k=0}^{\infty} \left(\frac{1+3k}{3^n}, \frac{2+3k}{3^n} \right) = A_{n-1} \cap \underbrace{\left(\mathbb{R} \setminus \bigcup_{k=0}^{\infty} \left(\frac{1+3k}{3^n}, \frac{2+3k}{3^n} \right) \right)}_{\text{complement of open, so closed}}.$$

Thus the intersection $C = \bigcap_{n=1}^{\infty} A_n$ of closed sets is closed, and C is bounded because it is contained in the bounded set $[0, 1]$, which means it is compact by Munkres's Theorem 27.3.

(c) This follows from induction: clearly it is true for A_0 , and if A_n consists of finitely many closed intervals of length 3^{-n} , then removing the middle third $((2a+b)/3, (a+2b)/3)$ of some closed interval $[a, b]$ leaves behind the closed intervals $[a, (2a+b)/3]$, $[(a+2b)/3, b]$ with a third of the length. Furthermore, since removing the middle third necessarily means that only interior points of each interval are removed and the endpoints of each interval stay as endpoints of some new intervals (as in the one-interval example $[a, b]$ above), one can never lose the endpoints via the middle-thirds-deletion process, so the endpoints are in C .

(d) For each $x \in C$ and $n \in \mathbb{N}$, since $x \in A_n$ it is contained in some interval of length 3^{-n} that is contained in A_n . The endpoints of this interval are in C by part (c) and are at most 3^{-n} away from x ; let x_n be one of these endpoints (if x is an endpoint of the interval then pick x_n to be the other endpoint). We have thus found a point $x_n \in C$ not equal to x such that $|x - x_n| < 3^{-n}$. This means that $\lim_{n \rightarrow \infty} x_n = x$, so any open set around x contains some (in fact cofinitely many) x_n (none of which are equal to x), so x is not an isolated point.

(e) Since C is a nonempty compact Hausdorff space with no isolated points, Munkres's Theorem 27.7 implies that C is uncountable. $\square$